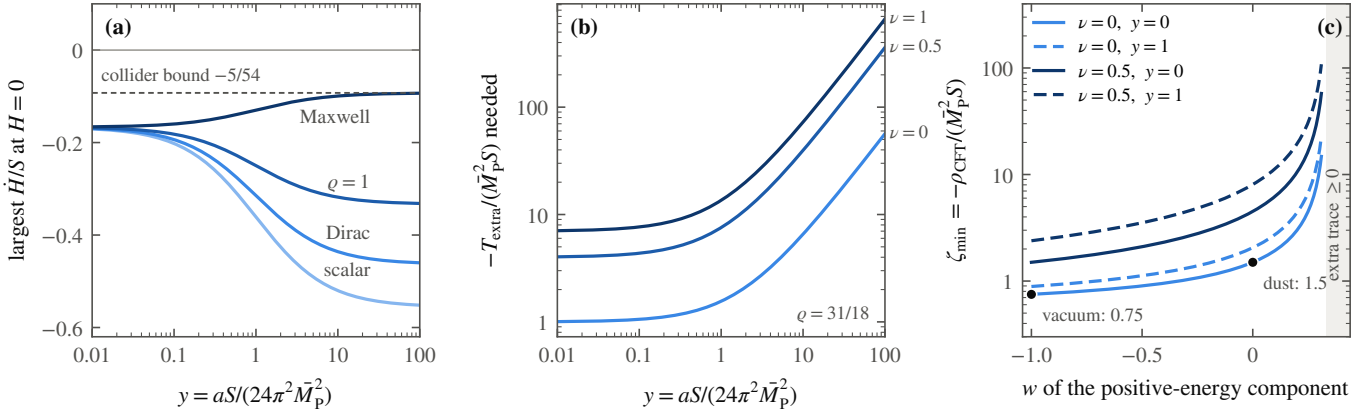
(a)
collider bound −5/54
Maxwell
ϱ = 1
Dirac
scalar
largest Ḣ/S at H = 0
y = aS/(24π²M̄²P)
(b)
ν = 1
ν = 0.5
ν = 0
ϱ = 31/18
−T_extra/(M²PS) needed
y = aS/(24π²M̄²P)
(c)
ν = 0, y = 0
ν = 0, y = 1
ν = 0.5, y = 0
ν = 0.5, y = 1
ζmin = −ρCFT/(M̄²PS)
vacuum: 0.75
dust: 1.5
extra trace ≥ 0
w of the positive-energy component